Home / Grade 11 / Geometry / Quarter 2

Upper and lower bounds in measurement

Cambridge insert: every measurement is an interval, and every calculation from it is an interval too.

LESSON 24 | 1 × 40 MIN | GEOMETRY 11, §11 (EXTENSION) | IGCSE E13.3

LESSON CLOCK

8' | 12' | 14' | 8' | 3'

A measurement is an interval	8 min
The four operations	12 min
Areas and volumes	14 min
How many figures	8 min
Homework	3 min

LEARNING OBJECTIVES

- Write the upper and lower bound of a rounded measurement.
- Find the bounds of a sum, difference, product and quotient.
- Apply bounds to the area and volume of a solid.
- Decide how many figures an answer can honestly carry.

TEXTBOOK

National	pp. 119–122
Cambridge	Core & Extended, pp. 44–48
Workbook	IGCSE Exercise 13.3

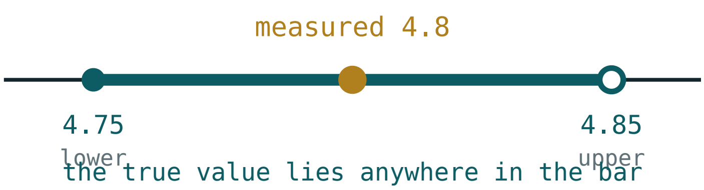
01 Explanation

A measurement is an interval

HALF THE ROUNDING UNIT, EACH WAY

A length given as 8 cm to the nearest cm lies in $7.5 \le \ell < 8.5$. To the nearest 0.1 cm, 8.0 means $7.95 \le \ell < 8.05$.

The lower bound is included and the upper is not — but at this level both are written as ≤, and the distinction is only mentioned.



The interval a rounded value stands for. Half a unit each side.

The four operations

QUANTITY	GREATEST WHEN	LEAST WHEN
$a + b$	both are greatest	both are least
$a - b$	a greatest, b least	a least, b greatest
$a \times b$	both greatest	both least
$\frac{a}{b}$	a greatest, b least	a least, b greatest

SUBTRACTION AND DIVISION SWAP THE BOUNDS

To make a difference as large as possible, take the largest top and the **smallest** bottom. Using the upper bound of both is the standard error.

Areas and volumes

A cuboid measured as $6 \times 8 \times 10$ cm, each to the nearest cm, has

$$V\_max = 6.5 \times 8.5 \times 10.5 = 580.125 \; V\_min = 5.5 \times 7.5 \times 9.5 = 391.875$$

The stated volume is 480 cm³, but the true value lies anywhere in a range of nearly 190 cm³. Small uncertainties in three lengths compound sharply in the product.

WHY THIS INSERT SITS HERE

Every surface-area and volume formula of this quarter multiplies two or three measured lengths. Bounds are how a real answer is stated, and the reason a measured result is never quoted to eight figures.

How many figures

Quote the answer only to the digits that agree between the upper and lower bounds. For the cuboid: $V\_min = 391.9$, $V\_max = 580.1$ — not even the first digit agrees, so the honest statement is “between about 390 and 580 cm³”.

Measure the same box to the nearest millimetre and the bounds become 478.5 and 481.5: now “480 cm³” is justified.

02 Worked examples

EXAMPLE 1 **A length is 12.4 cm to 1 d.p. Give its bounds.**

1. Rounding unit 0.1; half is 0.05.
2. $12.35 \le \ell \le 12.45$

ANSWER

12.35 and 12.45 cm

EXAMPLE 2 **A rectangle is 7 cm by 4 cm, each to the nearest cm. Find the bounds of its area.**

1. $6.5 \le a \le 7.5$, $3.5 \le b \le 4.5$.
2. $A_{min} = 6.5 \times 3.5 = 22.75$
3. $A_{max} = 7.5 \times 4.5 = 33.75$

ANSWER

22.75 to 33.75 cm²

EXAMPLE 3

A journey of *150* km (to the nearest 10 km) takes *2.5* h (to the nearest 0.1 h). Find the greatest possible speed.

1. Distance greatest: *155*.
2. Time least: *2.45*.
3. $\frac{155}{2.45} \approx 63.3$

ANSWER

≈ 63.3 km/h

03 Interactive model

Measure the same object with a ruler and with a tape and compare the intervals.

Bounds

8 cm to the nearest cm has lower bound:

- 7
- 7.5
- 7.9
- 8

4.60 to 2 d.p. has upper bound:

- 4.61
- 4.605
- 4.65
- 4.7

To maximise $a - b$ take:

- both maxima
- both minima
- a max, b min
- a min, b max

To maximise $\frac{a}{b}$ take:

- both maxima
- a max, b min
- both minima
- a min, b max

A 7×4 rectangle (nearest cm) has greatest area:

- 28
- 30
- 33.75
- 35

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Upper bound	Yuqori chegara	Верхняя граница
Lower bound	Quyi chegara	Нижняя граница
Rounded to the nearest	Yaxlitlangan	Округлено до
Degree of accuracy	Aniqlik darajasi	Степень точности
Interval of values	Qiymatlar oralig‘i	Промежуток значений
Maximum value	Eng katta qiymat	Максимальное значение
Minimum value	Eng kichik qiymat	Минимальное значение
Significant figures	Muhim raqamlar	Значащие цифры

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

20 m to the nearest 10 m lies in:

- $19–21$
- $15–25$
- $19.5–20.5$
- $10–30$

A measurement is best described as:

- an exact number
- an interval
- a guess
- a fraction

Bounds of a volume from three measured lengths:

- add
- multiply the corresponding bounds
- stay the same
- halve

An answer should be quoted to:

- as many figures as the calculator gives
- the figures the bounds agree on
- two figures always
- one figure

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Bounds of 9 cm to the nearest cm

   ANSWER 8.5 and 9.5

2. Bounds of 4.2 to 1 d.p.

   ANSWER 4.15 and 4.25

3. Bounds of 300 to the nearest 100

   ANSWER 250 and 350

4. Bounds of 12.40 to 2 d.p.

   ANSWER 12.395 and 12.405

5. Greatest $a + b$ for $a = 5, b = 3$ (nearest 1)

   ANSWER 9

6. Least $a + b$ for the same

   ANSWER 7

7. Greatest $a - b$ for the same

   ANSWER 3

Medium · the standard question

7 PROBLEMS

1. Rectangle 7×4 (nearest cm): area bounds

   ANSWER 22.75 to 33.75

2. Cuboid $6 \times 8 \times 10$ (nearest cm): volume bounds

   ANSWER 391.875 to 580.125

3. Speed: 150 km (nearest 10), 2.5 h (nearest 0.1). Greatest speed

   ANSWER ≈ 63.3 km/h

4. Same: least speed

   ANSWER ≈ 56.9 km/h

5. A square of side 5.0 cm (1 d.p.): area bounds

   ANSWER 24.5025 to 25.5025

6. A cube of edge 4 cm (nearest cm): volume bounds

   ANSWER 42.875 to 91.125

7. To how many figures can that volume be quoted?

   ANSWER none — not even the first digit agrees

Hard · stretch and reason

7 PROBLEMS

1. A circle of radius 6.0 cm (1 d.p.): area bounds

   ANSWER ≈ 111.2 to ≈ 115.0

2. A cylinder $r = 3, h = 10$ (nearest 1): volume bounds

   ANSWER ≈ 205.6 to ≈ 405.7

3. A density is $\frac{m}{V}$ with $m = 250$ g (nearest 10) and $V = 50$ cm³ (nearest 5). Greatest density

   ANSWER ≈ 5.42 g/cm³

4. Same: least density

   ANSWER ≈ 4.62 g/cm³

5. A rectangle has area 40 cm² and length 8 cm, both to the nearest unit. Greatest width

   ANSWER ≈ 5.4 cm

6. Explain why bounds widen faster for a volume than for a length

   ANSWER Three uncertainties multiply

7. A pipe is 2.0 m long to 1 d.p.; how many fit in 50 m to the nearest metre?

   ANSWER at least 24, at most 25

07 Homework

Homework — 4 tasks

Always state both bounds, and say which extreme you used and why.

1. Write the bounds of 15 cm (nearest cm), 3.7 m (1 d.p.) and 600 g (nearest 100 g).
2. A cuboid measures $4 \times 6 \times 9$ cm, each to the nearest cm. Find the bounds of its volume.
3. A car travels 80 km (nearest 10 km) in 1.5 h (nearest 0.1 h). Find the greatest and least possible speeds.

4. Explain, with the cuboid of task 2, why the answer cannot honestly be quoted to three figures.